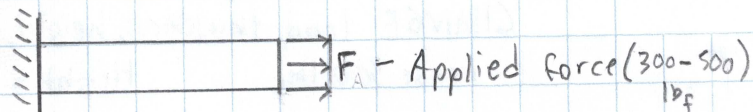


A3:

Given:

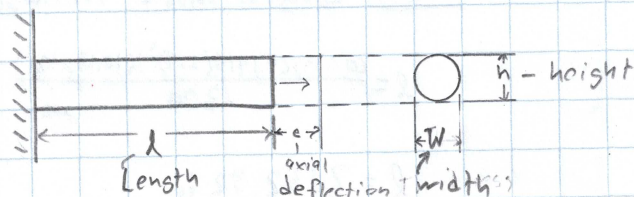
$$e(\text{inches}) = \frac{F(\text{lbs}) \cdot L(\text{in})}{A(\text{in}^2) \cdot E\left(\frac{\text{lbs}}{\text{in}^2}\right)}$$



F is a distributed load "between
 $300 \text{ lbs} < F < 500 \text{ lbs}$.

Maximum axial deflection (e)
 is 0.009 inches.

The bar's material is Aluminum,
 $(8.5 \times 10^6 \text{ psi} < E < 11.5 \times 10^6 \text{ psi})$.



Choices $F_A = 500 \text{ lbs}$ $E = 11.5 \times 10^6 \text{ psi}$

Width = 0.500 in ($r = 0.250 \text{ in}$)

$$e = \frac{FL}{AE} \rightarrow L = \frac{e \cdot A \cdot E}{F} \rightarrow L = \frac{(0.009) \left(\pi \left(\frac{w}{2}\right)^2\right) \cdot (11.5 \times 10^6)}{500}$$

$$= \frac{(0.009) (3.1415 (0.250)^2) (11.5 \times 10^6)}{500}$$

$$L = 40.6432 \text{ inches}$$

Changing Decision: $E = 8.5 \times 10^6$

$$L = \frac{(0.009) (3.1415 (0.250)^2) (8.5 \times 10^6)}{500}$$

$$L = 30.0406 \text{ in}$$

Changing Decision: $E = 10.15 \times 10^6 \text{ psi}$

$$L = \frac{(0.009) (3.1415 (0.250)^2) (10.15 \times 10^6)}{500}$$

$$L = 35.8720 \text{ in}$$

"Displacement" (From Solidworks) = 0.2284 mm

$$0.2284 \text{ mm} \times \left(\frac{1 \text{ in.}}{25.4 \text{ mm}}\right) = 0.00899 \text{ in.}$$